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EEE 202-1

LAB-03: Passive and Active Band-Pass Filters

Introduction:

The goal of this assignment is to design and analyze passive and active filters by utilizing phasor analysis. For the passive part, we will construct a third-order Butterworth low-pass filter. An inductor that has been assigned through a document and it's values will be used in conjunction with two capacitors whose values will be determined through the resonant and cutoff frequency of the filter. For the active part, an OPAMP will be used with capacitors and resistors to create a bandpass gain multiple feedback active filter with a center frequency and bandwidth which has also been assigned through a document. The behaviors of both filters' gain will be analyzed as a function of the input frequency. After the analysis has been conducted for both filters individually, they will be simulated using LTSpice to gather the plot for gain in decibels versus frequency to analyze overall behavior and to compare our derived results. Finally, the designs will be implemented in real life to compare the analytical and simulative results with real-life examples.

Analysis:

For this lab assignment, no specific derivation is required as all derivations are listed in the upcoming preliminary sections for the passive and active filters. However, the assigned component values and frequencies for the passive (Figure A) and active (Figure B) filters will be noted here.

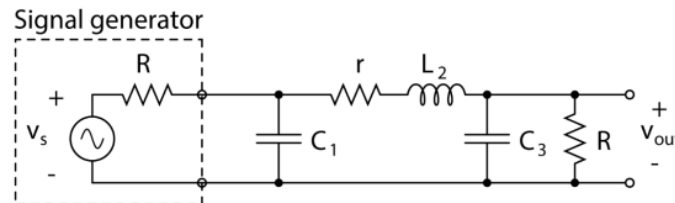


Figure A: Schematic of the passive third-order low-pass Butterworth filter

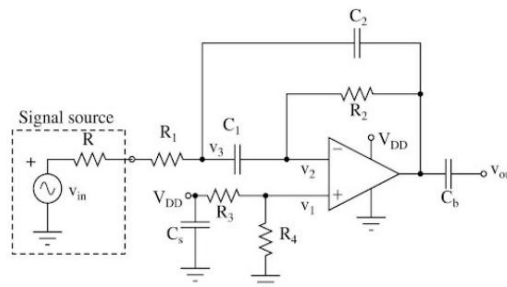


Figure B: Schematic of the active band-pass filter

In the Lab 3 Component Assignment document, we were assigned the inductor whose part number was “SLOT TEN-2-09-B”. This specific inductor has a listed minimum, maximum and nominal inductance of $16\mu H$, $35\mu H$ and $25.3\mu H$ respectively. Additionally, it’s quality factor is given to be 40 at a test frequency of 2.5 MHz. Using these values, we can find the inductor’s series resistance by inputting the given values into the equation $r = \frac{2\pi f_T L_{nom}}{Q}$ to find that it’s value is approximately 9.9353 Ohms.

For the upcoming active filter, by looking at our assigned frequency by taking the mod20 of our Bilkent ID (which yields 10), we have been given the design constraint that the center frequency is 3kHz and the bandwidth is 0.52kHz. As a remark, we can divide these terms to find the quality factor, which is approximately 5.7692.

Preliminary Part 1 (Capacitance and Frequency Calculation):

Using the given equations and the Lab 3 document ($f_0 = \frac{b_2 R}{2\pi L_2}$, $C_1 = \frac{b_1}{2\pi f_0 R}$, $C_3 = \frac{b_3}{2\pi f_0 R}$) and our derived values in the analysis part, we can directly find the 3-dB cutoff frequency and the capacitor values. In the Lab 3 document, R is assumed to be 50 Ohms and the coefficients b_1 , b_2 and b_3 are given to be 1, 2 and 1 respectively, which is in accordance to the parameters required to construct a Butterworth filter of the third order. Inputting the values in to the equations, we find f_0 to be approximately 629 kHz, and the capacitor values C_1 and C_3 to be equal and be equal to approximately 5.0606 nF.

Preliminary Part 2 (Transfer Function Plot):

The given third-order Butterworth low-pass filter’s transfer function was implemented in MATLAB and plotted (Figure C). It was observed that the plot shows the frequency for the 3-dB cutoff frequency to be 629 kHz, which matches our analytical result. Additionally, by comparing two points where one’s frequency is double the other, we can observe that the filter exhibits an -18.0718 dB/octave roll-off $\left(\frac{-72.0029 + 90.0609}{\log_2\left(\frac{19929600}{9970090}\right)} \right)$ after the cutoff frequency. This yields an error of 0.3989%, which can be caused by floating point errors.

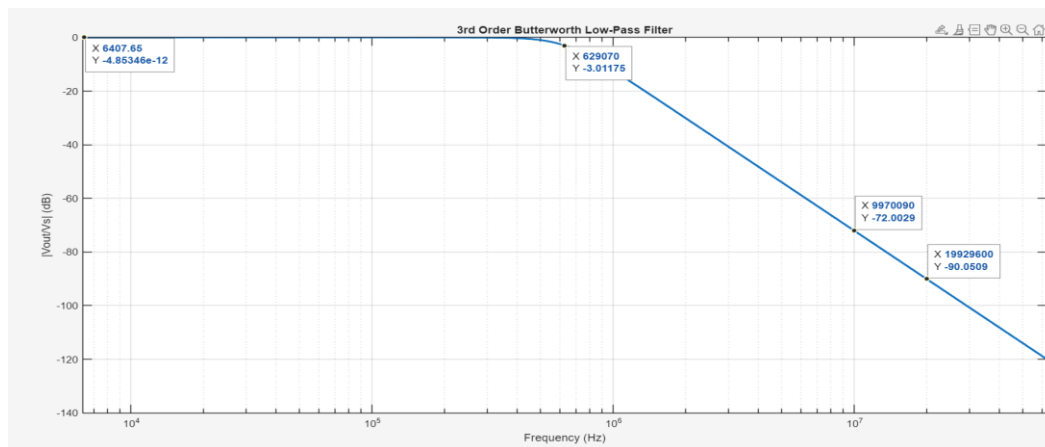


Figure C: MATLAB plot of the low-pass filter with cutoff frequency and bode measuring points highlighted (Code in Appendix)

Preliminary Part 3 (Low-Pass Filter Simulation):

By using the given parameters and the derived values, the low-pass filter circuit in Figure A was made in LTSpice. In order to view the gain of the filter versus frequency, an AC analysis was performed and the output voltage was probed. The plot was cursorred to view the insertion loss at $f_0/10$ and the 3-dB cutoff frequency (Figure D). Additionally, two frequency points after the 3-dB cutoff – where one's frequency the double the other – were cursorred to measure the per octave roll-off in dB (Figure E).

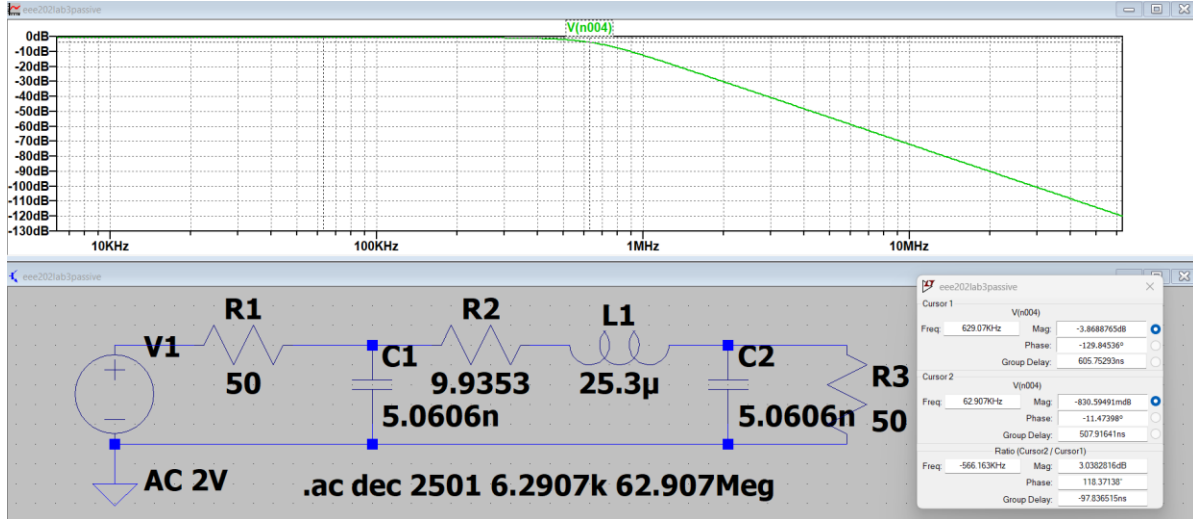


Figure D: LTSpice simulation of the low-pass filter with cutoff frequency and insertion loss points cursorred

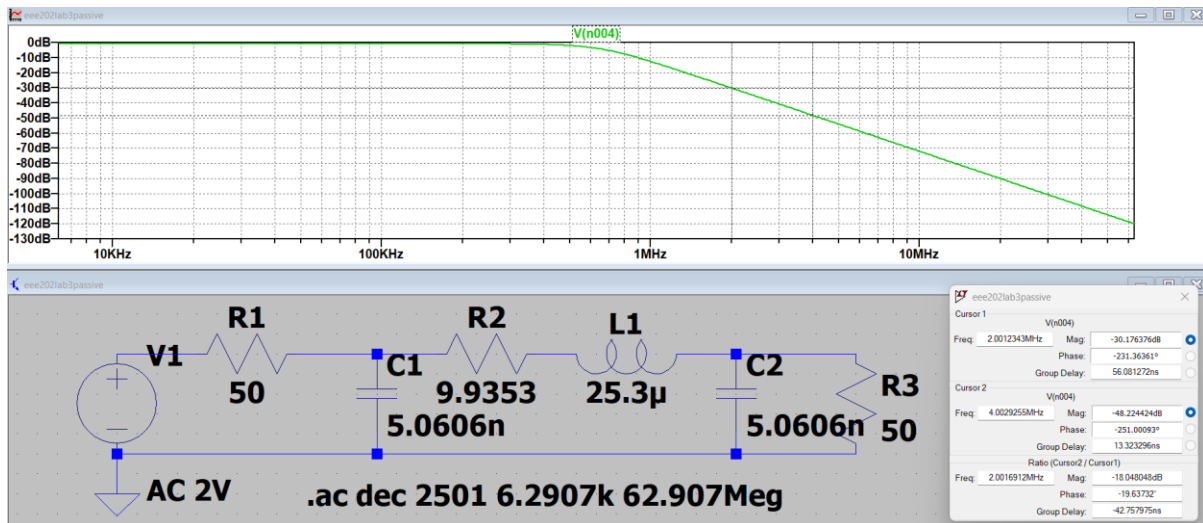


Figure E LTSpice simulation of the low-pass filter with bode measuring points cursorred

By analyzing the first figure, the insertion loss was found to be approximately -0.8306 dB and the cutoff frequency was found to be 629 kHz, which matched our expected result. Additionally, by analyzing the stopband drop-off, the roll-off was also found to be approximately -18.0447dB/octave $\left(\frac{-30.1764 + 48.2244}{\log_2\left(\frac{4.0029}{2.0012}\right)}\right)$.

A slight error of 0.2483% was observed when comparing the simulation to the ideal result, however, this could be caused by the software handling floating point numbers differently, causing an extremely small error for both the roll-off and the cut-off frequency. Another cause for discrepancy could be due to the inductor resistance interfering with the overall gain and its quality.

Preliminary Part 4/5 (Active Bandpass Filter Design and Derivations):

For this part, the transfer function will be derived for the active bandpass filter in Figure B. Additionally, the equations governing certain parameters will be proven through KCL and the transfer function. For this case and related derivations, all components will be assumed to exhibit ideal model behaviors. For example, the OPAMP will have an infinite gain and the reactive components will not have any resistive components embedded in them. For the transfer function, we can apply KCL at nodes v_2 and v_3 by only looking at AC sources and assuming the DC input components have a voltage of zero. The node equations at these nodes can be written initially as:

$$\frac{v_3 - v_{in}}{R'_1} + \frac{v_3 - v_{out}}{Z_2} + \frac{v_3 - v_2}{Z_1} = 0 \quad (1)$$

$$\frac{v_2 - v_3}{Z_1} + \frac{v_2 - v_{out}}{R_2} = 0 \quad (2)$$

where $R'_1 = R_1 + R$ and Z_1 and Z_2 are the impedances of C_1 and C_2 respectively. It should be noted that capacitors exhibit an impedance relation of $Z = \frac{1}{j\omega C}$ where Z is the impedance, ω is the angular frequency and C is the capacitance. For the given design, we assume C_1 and C_2 to be equal and their common impedances and capacitances will be denoted with Z and C respectively. Additionally, for this small-signal analysis, we can see that v_1 is equal to zero as we have killed the DC supply that was connected to it. Due to the ideal OPAMP relation of $v_1 = v_2$, we can say that the voltage at node v_2 is also equal to zero. Inserting common capacitance and v_2 voltage into equation (2), we have:

$$\frac{v_3}{Z} + \frac{v_{out}}{R_2} = 0 \rightarrow v_3 = \frac{-Zv_{out}}{R_2} \quad (3)$$

where $Z = \frac{1}{j\omega C}$. By utilizing this relation and the fact that $v_2 = 0$ in equation (1), we can further write:

$$\frac{-Zv_{out} - R_2v_{in}}{R'_1R_2} + \frac{-Zv_{out} - R_2v_{out}}{ZR_2} + \frac{-v_{out}}{R_2} = 0 \rightarrow (Z^2 + 2ZR'_1 + R'_1R_2)v_{out} + R_2Zv_{in} = 0 \quad (4)$$

We can rearrange equation (4) to get the transfer function in the form $\frac{v_{out}}{v_{in}}$, which can be written as:

$$\frac{v_{out}}{v_{in}} = -\frac{R_2Z}{Z^2 + 2ZR'_1 + R'_1R_2} \quad (5)$$

Finally, we can insert the relation $Z = \frac{1}{j\omega C}$ into equation (5) to find $H(\omega)$. Therefore, the transfer function becomes:

$$H(\omega) = \frac{v_{out}}{v_{in}} = -\frac{j\omega CR_2}{1 + 2j\omega R'_1C - \omega^2 C^2 R'_1R_2} \quad (6)$$

For arbitrary C , R_2 , R'_1 and ω for the design in Figure B. From this, we can calculate gain and center frequency for our active bandpass filter. Before that, component values will be determined.

First, it was determined for this specific case that the filter's center frequency is 3 kHz, it's bandwidth is 0.52 kHz and it's quality factor is approximately 5.7692 (see Analysis). From the Lab 3 document, R_1 has to have a value over 500 Ohms. Since by the given relation of $Q \approx \sqrt{\frac{R_2}{4R'_1}}$ is present in the Lab 3

document, and the need to abide by values adhering to the standard values, R_1 was selected to be 560 Ohms and R_2 was selected to be 82 kOhms. This is due to the selected values yielding a close approximation of the quality factor. With the selected resistance values inserted into the equation, we get the approximate value of $\sqrt{\frac{82000}{4(560+50)}} = 5.7971$, which has an error of 0.4813% when compared to the ideal quality factor. From this, we can calculate the capacitor capacitances by rearranging $f_0 = \frac{1}{2\pi C_1 \sqrt{R'_1 R_2}}$

to $C_1 = \frac{1}{2\pi f_0 \sqrt{R'_1 R_2}}$. With our determined resistances and the assigned frequency of 3 kHz, the capacitance C_1 was determined to be 7.5011 nF. Since this exact capacitance doesn't exist, we will use one 6.8 nF capacitor in parallel with a 680 pF capacitor to get an equivalent capacitance of 7.48 nF. This value has an error of 0.2813% when compared to the ideal value. It should be noted, as per the design parameters, that the found capacitance is $C = C_1 = C_2$. Which means we have determined both capacitor values for our circuit. The DC voltage V_{DD} had no specific constraint, so it was assigned the value of 12 Volts. The resistance values R_3 and R_4 had no conditions other than being larger than 1 kOhm, so both of their values were chosen to be 10 kOhms. Finally, the bypass and DC block capacitors C_s and C_b were arbitrarily chosen to be 100 nF and 10 μF respectively. By determining these values, we have designed an active bandpass filter with a center frequency of 3 KHz and a bandwidth of 0.52 kHz.

Now, the center frequency and gain derivations will be made through the transfer function. First, it should be noted that for bandpass filters, the maximum gain occurs at the center frequency. Additionally, the center frequency for linear circuits is the same as the resonant frequency. Since resonance occurs when the transfer function exhibits strictly real terms with no imaginary value, we can try to derive a value for ω for which the transfer function is purely real. We can achieve this by letting the real part of the denominator of equation (6) cancel out and equal zero so that there are only imaginary terms that divide each other to output a purely real relation. By doing this, we can derive ω_0 by taking the positive solution:

$$1 - \omega^2 C^2 R'_1 R_2 = 0 \rightarrow \omega_0 = \frac{1}{C \sqrt{R'_1 R_2}} \rightarrow f_0 = \frac{1}{2\pi C \sqrt{R'_1 R_2}} \quad (7)$$

Since in our design, we set the capacitances $C = C_1 = C_2$, we have proven that the center frequency f_0 is equal to the given relation of $\frac{1}{2\pi C_1 \sqrt{R'_1 R_2}}$. Since this active bandpass filter yields the maximum gain at the

center frequency, we can insert the found angular frequency in equation (7) into our transfer function in equation (6) to find the maximum gain of the circuit. As we have shown, the real part of the denominator cancels out, hence, we can simplify the transfer equation and insert the center frequency to get:

$$H(\omega_0) = -\frac{j\omega_0 C R_2}{2j\omega_0 R'_1 C} = -\frac{R_2}{2R'_1} \quad (8)$$

by taking the absolute value of this expression, we find the maximum gain of the design to be:

$$|A_v| = \frac{R_2}{2R'_1} \quad (9)$$

Which is the same as the given relation in the Lab 3 document. Now, we can simulate our design.

Preliminary Part 6 (Active Bandpass Filter Simulation):

In order to test our design, the circuit in Figure B was formed using LTSpice. The text file LM358 was gathered from Moodle and saved onto the project directory for LTSpice and included in the simulation. The derived component values from the Analysis and Preliminary Work 4/5 sections were used in the circuit and an AC analysis over a frequency range of $0.1f_0$ to $10f_0$ was performed and plotted (Figure F).

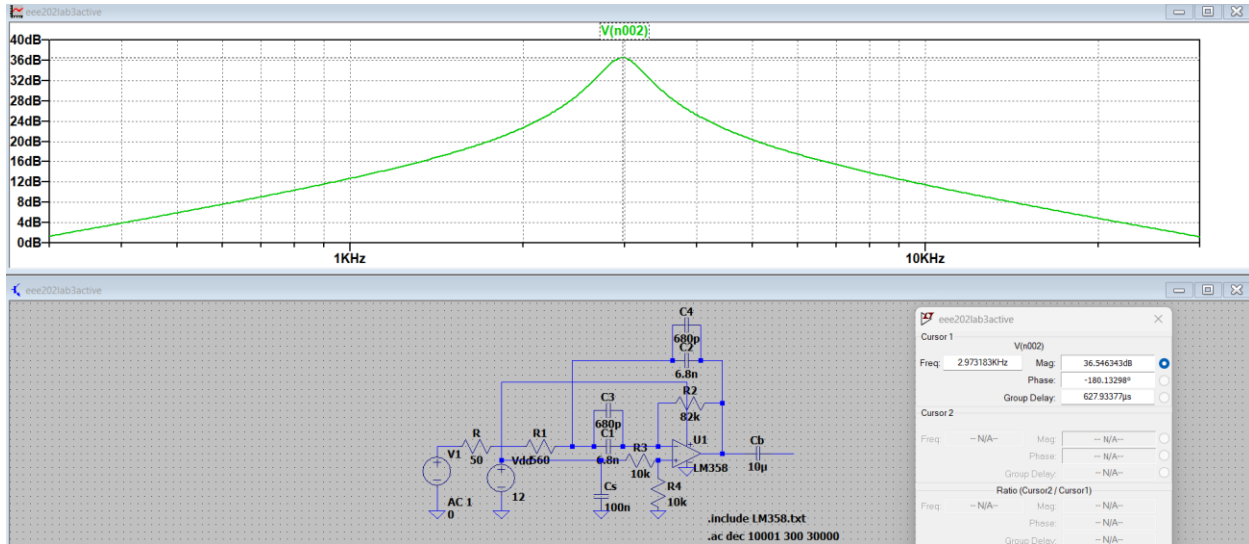


Figure F: LTSpice simulation of the bandpass filter with maximum gain and it's frequency censored

By inspecting the given plot, the maximum gain of the filter was observed to be 36.5463 dB at the center frequency of 2.9732 kHz. Since we expected a gain of 36.5491 dB by equation (9) ($|A_v| = \frac{82000}{2(610)}$, $20 \log(A_v) = 36.5491 \text{ dB}$) at a center frequency of 3 kHz (see Analysis), an error of 0.0197% and 0.8933% were observed for the gain and center frequency respectively. These could be caused by us getting an approximation of the quality factor Q and it having an effect on the gain. Additionally, since our capacitor values were also approximate, the center frequency can be judged to be less than expected. Also, rounding point errors by LTSpice can also be shown as a cause of error. After these derivations, the bandwidth of our design was found by finding the frequencies wherein the gain was 3 dB less than the maximum (Figure G).

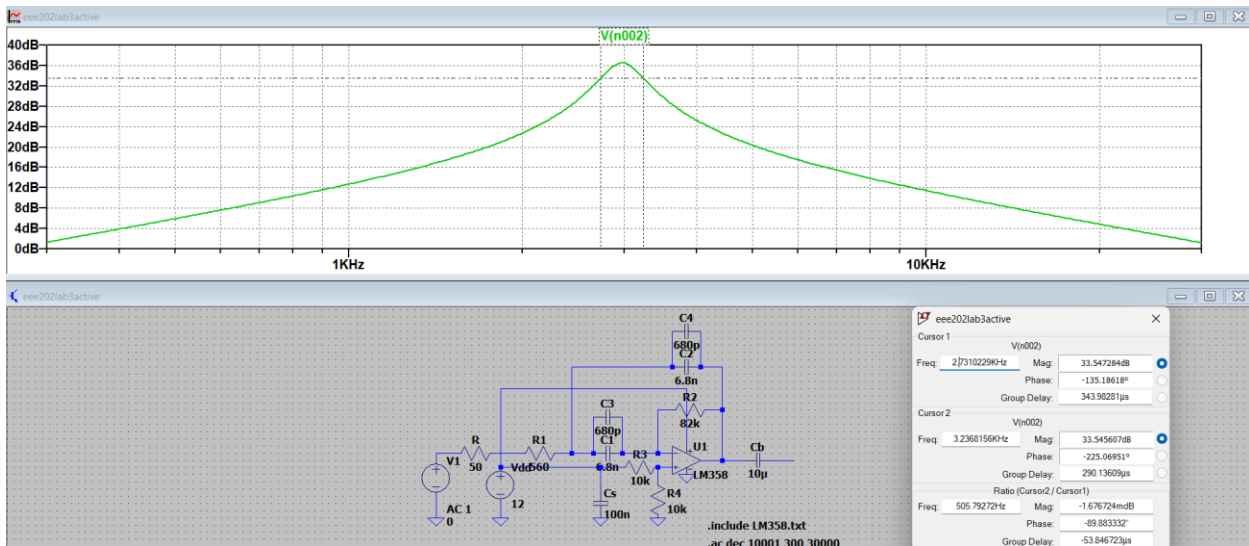


Figure G: LTSpice simulation of the bandpass filter with its cutoff frequencies censored

By looking at the position of the cursors, we can observe the bandwidth to be approximately 0.5058 kHz. Since we expected a bandwidth of 0.52 kHz per the Analysis section, an error of 2.7308% was observed. This can be caused by the OPAMP being unideal and us having a non-exact quality factor per the standard value capacitor and resistor values. Finally, the dB roll-off for high and low frequencies were to be calculated by observing the plot at the maximum measurable frequencies for the plot (Figure H, Figure I).

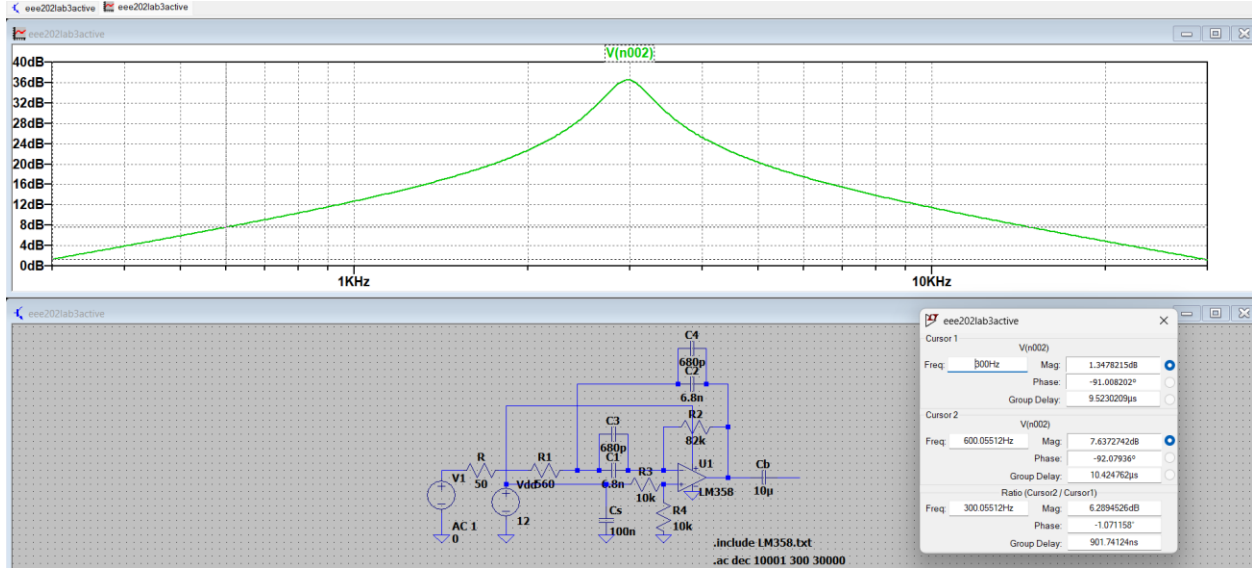


Figure H: LTSpice simulation of the bandpass filter with its low frequency gains censored

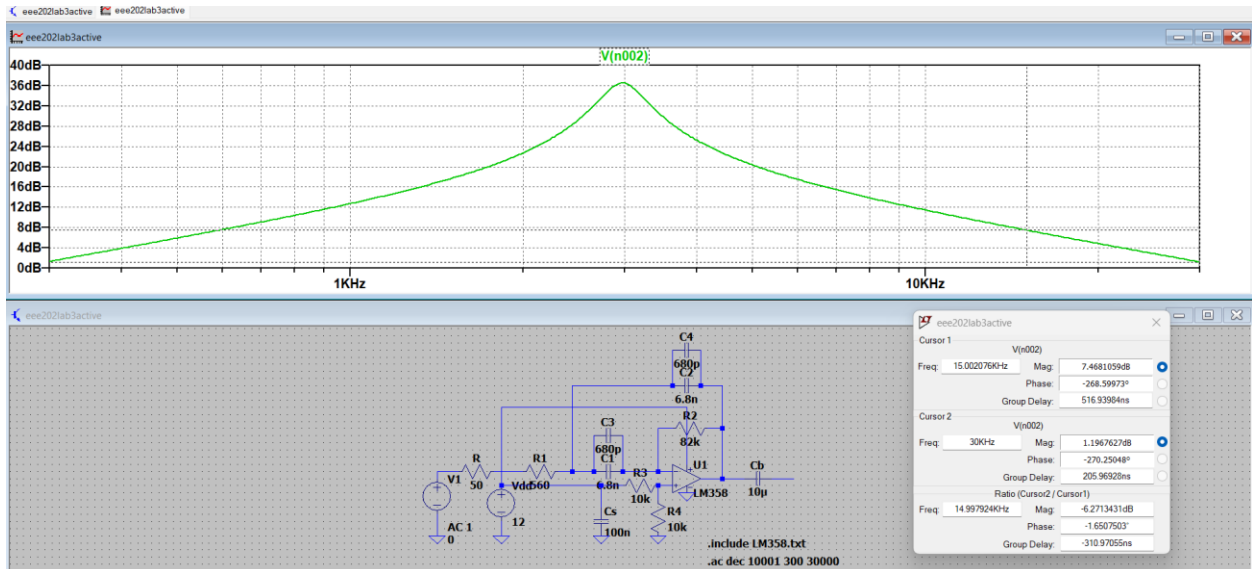


Figure I: LTSpice simulation of the bandpass filter with its high frequency gains censored

By inspecting the cursor points, the low and high frequency responses were found to be 6.2886 dB/octave and -6.2727 dB/octave respectively through the relation $\frac{dB_2 - dB_1}{\log_2(\frac{f_2}{f_1})}$ for both responses. Since we expected an

ideal response of ± 6 dB/octave, an error of 4.8100% and 4.5450% were calculated for the low and high frequencies respectively. This can be caused by the OPAMP not being ideal, standard component values, measuring at a non-sufficient range of frequencies and LTSpice having floating-point rounding errors.

By doing all of these design steps and derivations, we have gathered a strong basis for the hardware part of the assignment.

Hardware Implementation:

After the preliminary work section was done, the designs were to be implemented in real-life. First, the passive low-pass filter was to be made. In order to properly make the model, the assigned inductor's value was to be measured alongside with its quality factor. Using an LRC Meter, these values were observed (Figure 1.1). Using these results, the series resistance with the inductor could be found by additionally supplementing the test frequency into the formula as 100 KHz ($r = \frac{2\pi f_T L_{nom}}{Q}$). The series resistance r was more accurately found to be 3.0979 Ohms.



Figure 1.1: Measured inductance and quality factor of the given inductor

Using this information, the LTSpice model was updated for later error comparisons and measurements and the output voltage was plotted (Figure 1.2).

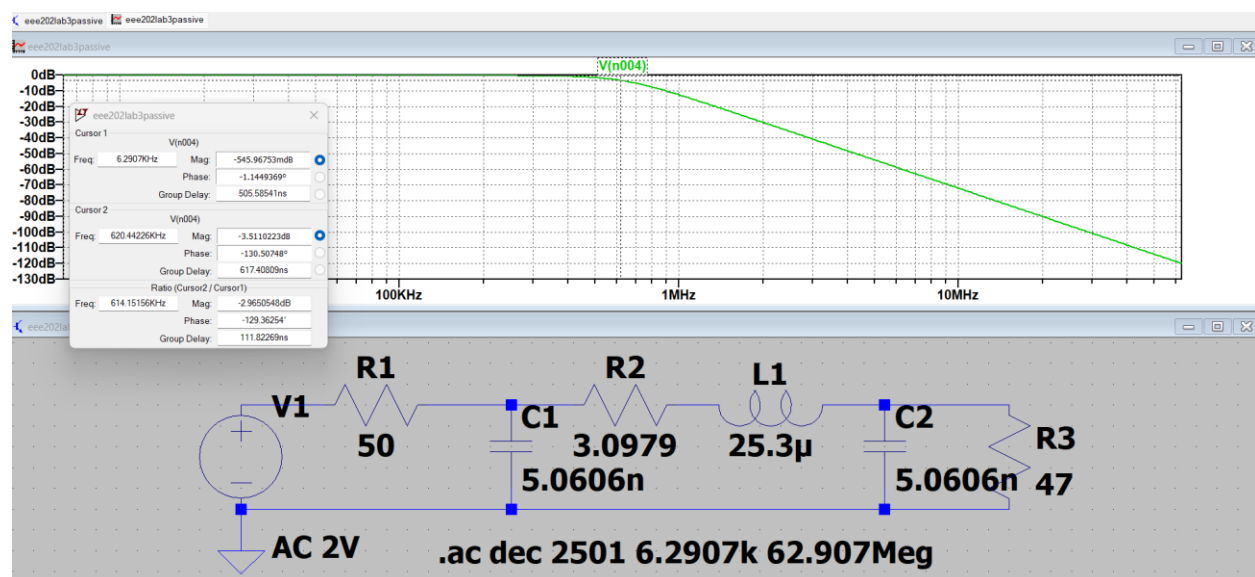


Figure 1.2: Updated passive filter plot with measured component values

From the graph, the new insertion loss and the cutoff frequency were found to be -0.5459 dB and 620 kHz respectively. These data points will be used when comparing the real-life values to the theoretical ones.

After these steps, the diagram in Figure A was formed on a breadboard with the right component values attached (Figure 1.3). The capacitances of the capacitors were measured with an LRC Meter and their values were found to be incredibly close to the ideal values (approximately 5.01 and 5.06 nF). One common ground node was used to minimize the unnecessary component values that could interfere with the voltage.

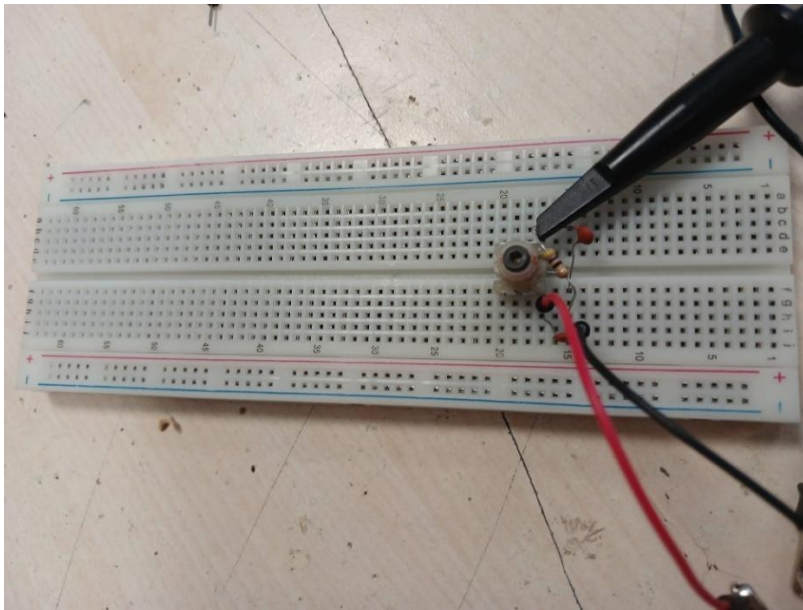


Figure 1.3: Implemented passive low-pass filter

As per the LTSpice model, the center frequency was determined to be 620 kHz and further measurements were made by using said frequency. Output voltage at each frequency point was measured using an oscilloscope probe while supplying 2 Volts to the circuit. Data entries at important values were photographed (Figure 1.4-1.7) and the voltages per frequency for each entry was also noted (Table 1) and plotted using the provided MATLAB code on Moodle (Figure 1.8).

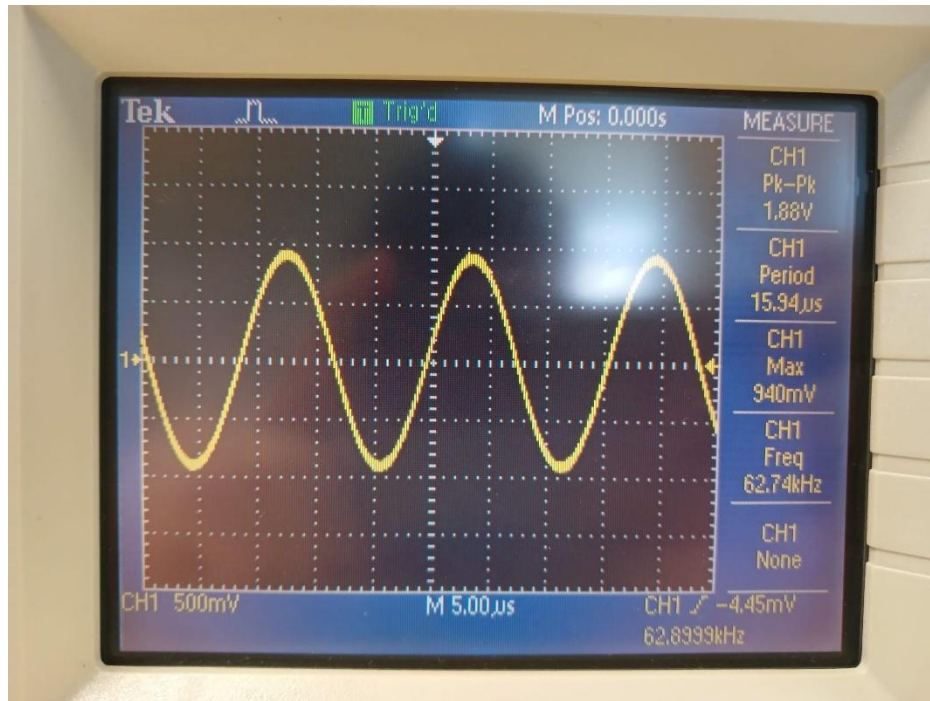


Figure 1.4: Output Voltage at $f_0/10$ and 2 Volt Input

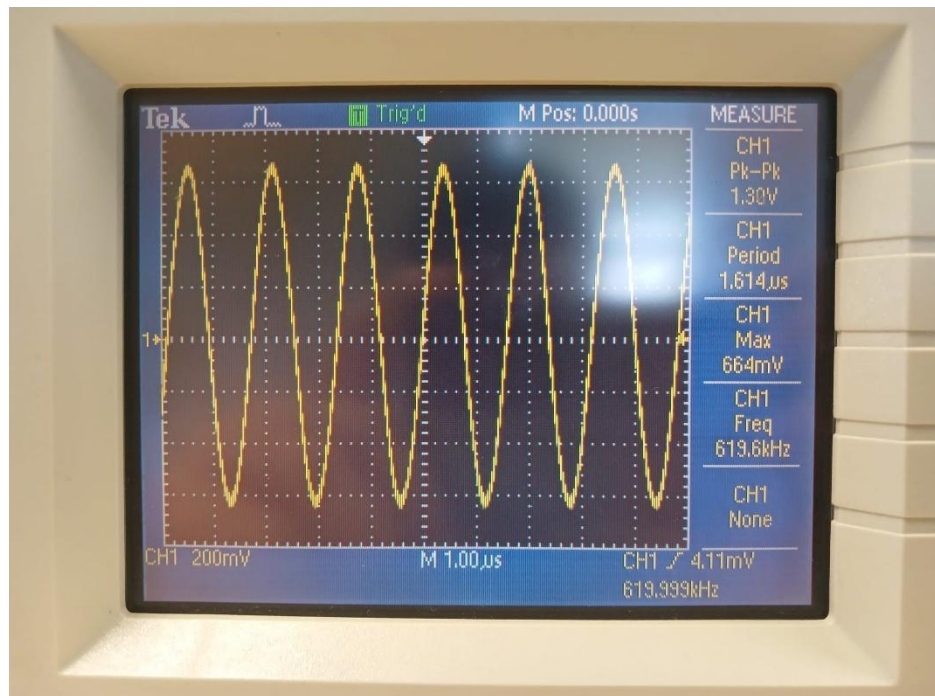


Figure 1.5: Output Voltage at f_0 and 2 Volt Input

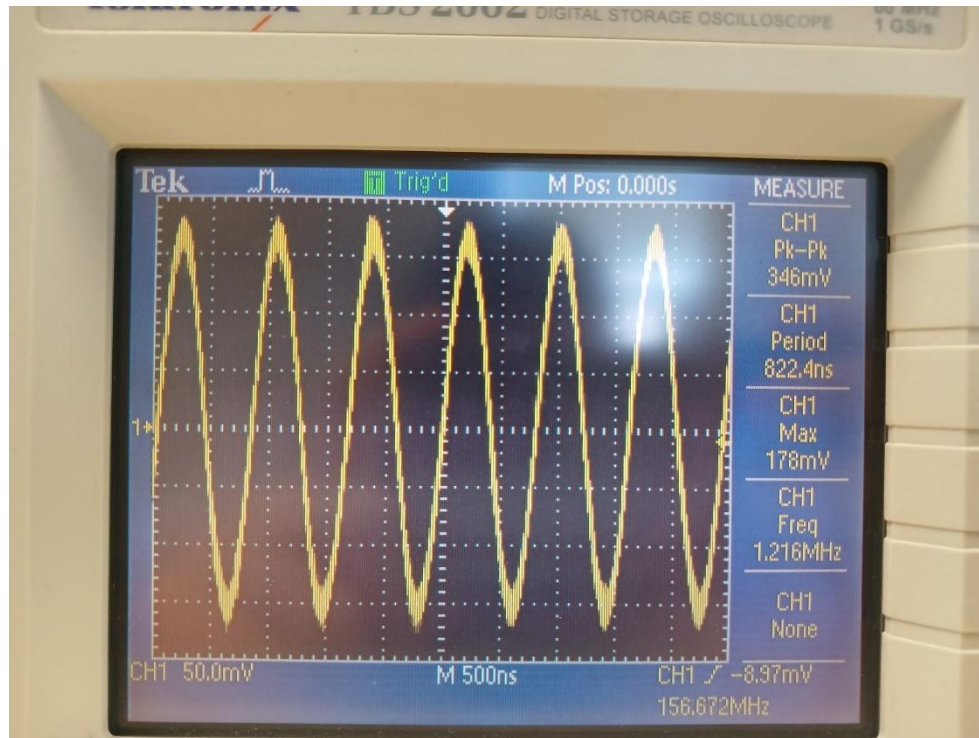


Figure 1.6: Output Voltage at $2f_0$ and 2 Volt Input

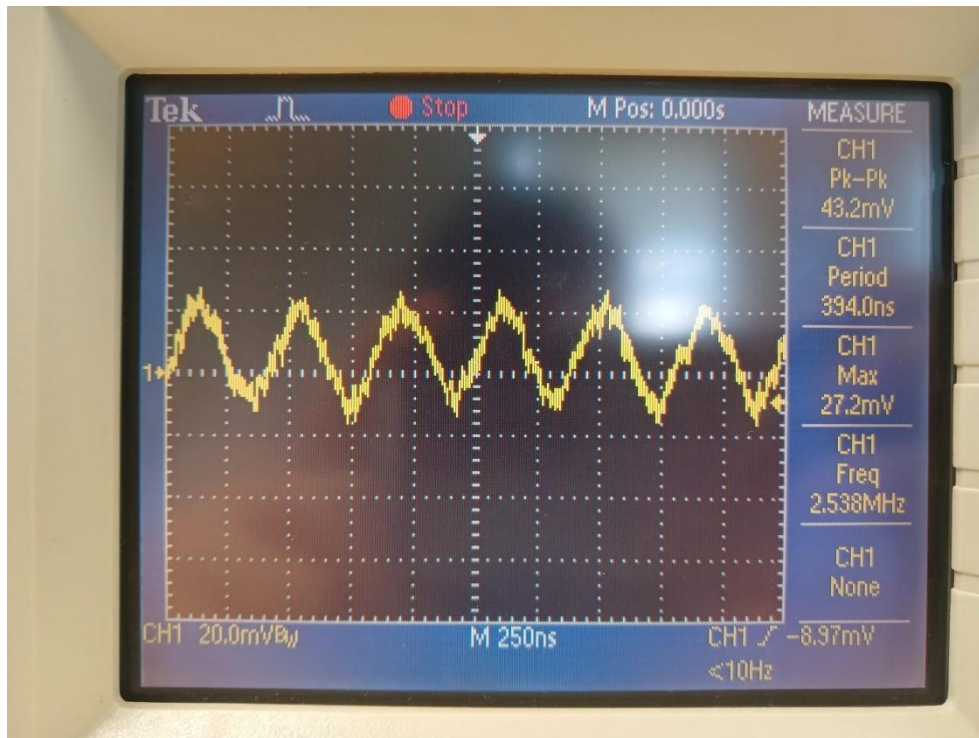


Figure 1.7: Output Voltage at $4f_0$ and 2 Volt Input

f_0	Freq (kHz)	v_{out} (V)	v_{out}/v_{in}	v_{out}/v_{in} (dB)
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0.1	62	1.88	0.94	-0.53744
0.2	124	1.84	0.92	-0.72424
0.4	248	1.76	0.88	-1.1103
0.8	496	1.64	0.82	-1.7237
1	620	1.3	0.65	-3.7417
1.2	744	0.9	0.45	-6.9357
1.4	868	0.68	0.34	-9.3704
1.6	992	0.528	0.264	-11.568
1.8	1116	0.416	0.208	-13.639
2	1240	0.346	0.172	-15.289
4	2480	0.0432	0.0216	-33.311
6	3720	0.204	0.102	-19.828
8	4960	0.236	0.118	-18.562
10	6200	0.29	0.145	-16.773

Table 1: Measured Data for the passive low-pass filter at given frequencies

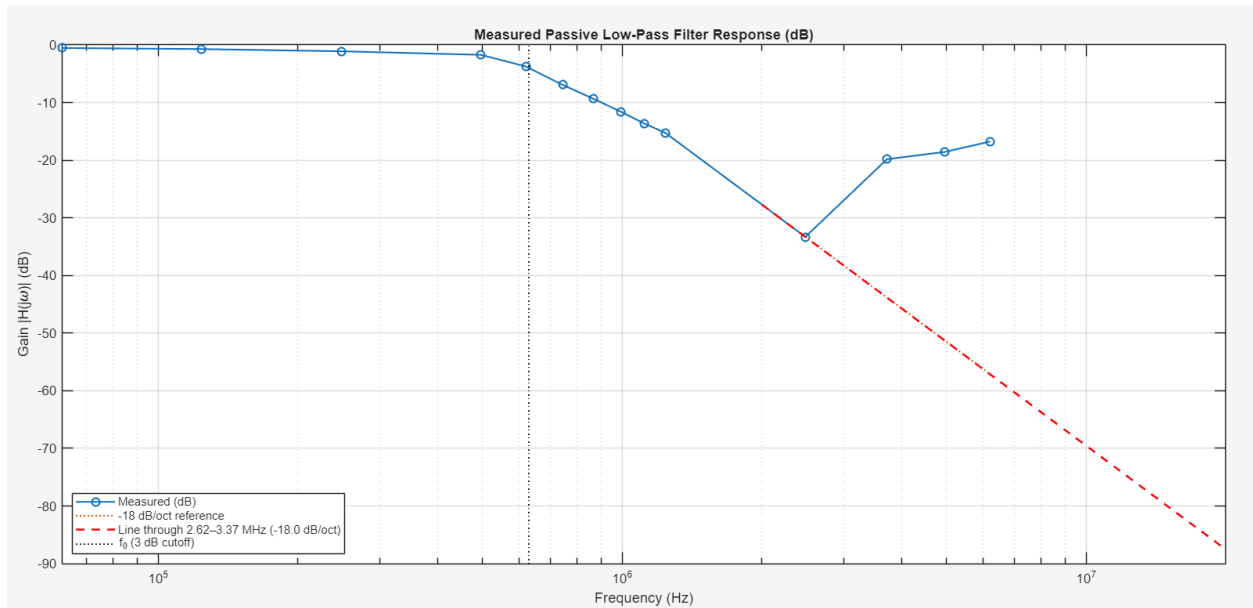


Figure 1.8: Measured gain versus frequency using the data from Table 1 (Code from Moodle)

Using these values, we were able to compare the real-life values to our simulations. First, the insertion loss was found to be -0.5374 dB when we expected -0.5460 dB from the updated LTSpice model, yielding an error of 1.5678%. The measured cutoff frequency was 620 kHz when we expected 629 kHz, yielding an error of 1.4308%. The gain differential between the insertion loss and the cutoff was expected to be -2.9566 dB according the updated LTSpice model, but was measured to be -3.2043 dB, yielding an error of 8.3779%. Due to the small nature of these errors, they can be attributed to the components not being ideal and human error when measuring.

A measurement that required special care when being calculated was the octave roll-off after the cutoff frequency. In the measurements at large frequencies, the plot heavily diverged from the expected LTSpice graph. As a sidenote, the most likely explanation for this phenomenon can be the capacitors behaving like inductors at very high frequencies as that is the expected behavior for real-life components. Due to this

fact, in order to calculate the roll-off slope, the two datapoints that are the closest to the LTSpice plot and decreasing points were selected for this calculation, which were $2f_0$ and $4f_0$ for this specific case. Using the relation of $\frac{dB_2 - dB_1}{\log_2(\frac{f_2}{f_1})}$ for the calculation at said frequencies and their respective gains, a roll-off per octave was found to be -18.022dB. By looking at the updated LTSpice plot at the same frequencies and using the same relation, a roll-off of -17.8602dB/octave was observed, yielding an error of 0.8980%. This small error can again be explained by component tolerances and human error whilst measuring. All errors for the passive low-pass filter were put on a table for easy reading (Table 2).

	Expected	Measured	Error
Insertion loss	-0.5460 dB	-0.5374 dB	1.5678%
Cut-off Frequency	629 kHz	620 kHz	1.4308%
Cut-off Gain Delta	-2.9566 dB	-3.2043 dB	8.3779%
Roll-off Slope	-17.8602dB/octave	-18.022dB/octave	0.8980%

Table 2: Errors for the measured vs. expected values for the passive low-pass filter

After all of these measurements and calculations were finished, the circuit was disassembled and the inductor was returned. Next up, the design for the active bandpass filter was to be tackled.

Using the design constraints given on Moodle and the derived values in the preliminary section, the active bandpass filter was to be designed. First the OPAMP IC was gathered from the lab with the required resistors and capacitors to have the required materials for the design. The connections were made in order to match the design in Figure F. The design was implemented on a breadboard and power was connected to the points of interest and the oscilloscope was connected to the output (Figure 2.1).

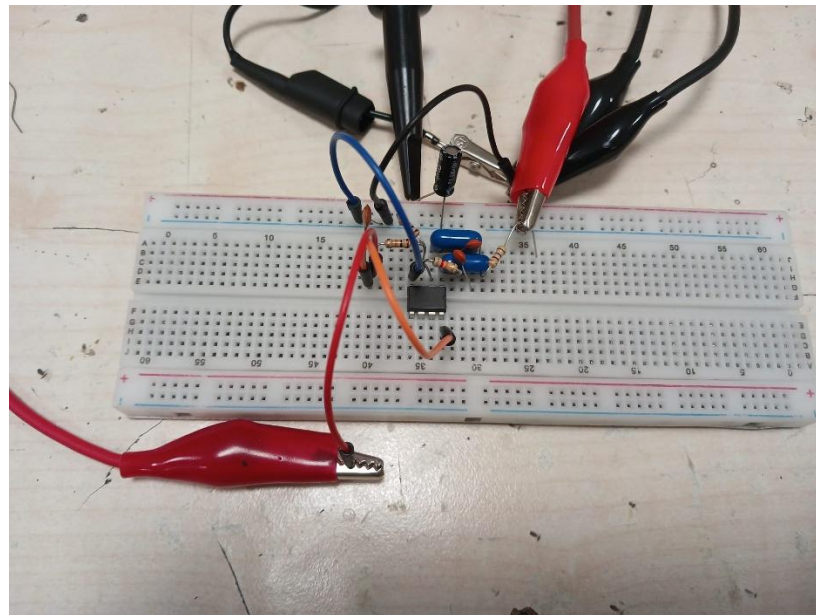


Figure 2.1: Implemented active bandpass filter

After the design was implemented, an input voltage of 20mV was applied as the input signal and the OPAMP biasing was supplied with 10V DC. After altering the frequency around the expected center frequency of 2.973 kHz, the real-life center frequency was found to be 3.08 kHz (for future reference, an

error of 3.5991%). After finding the center frequency to be this value, data entries at important values were photographed (Figure 2.2-2.6) and the voltages per frequency for each entry was also noted (Table 3) and plotted using the provided MATLAB code on Moodle (Figure 2.7).

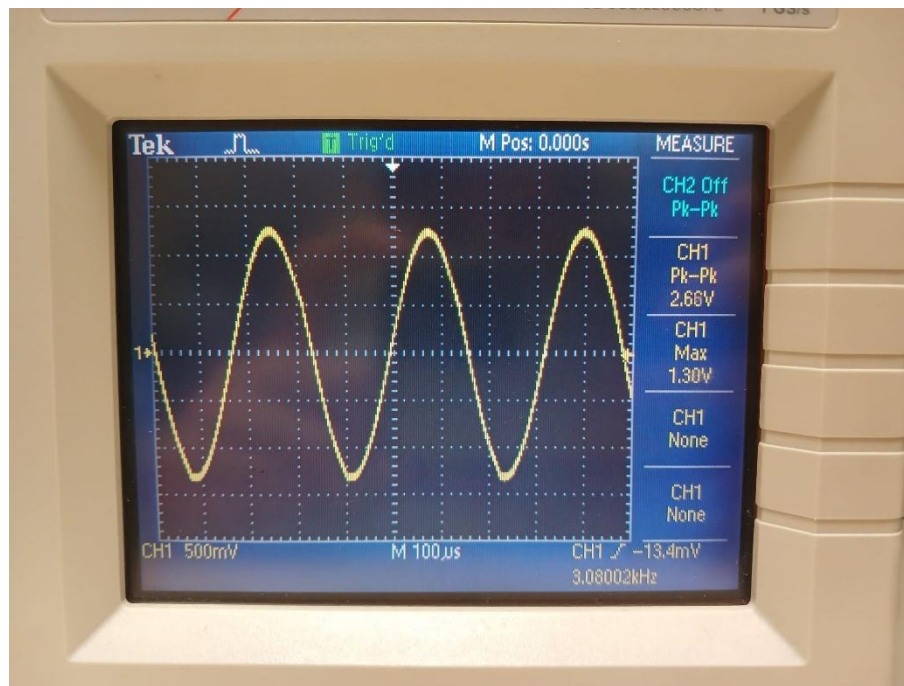


Figure 2.2: Output Voltage at f_0 with 20mV input

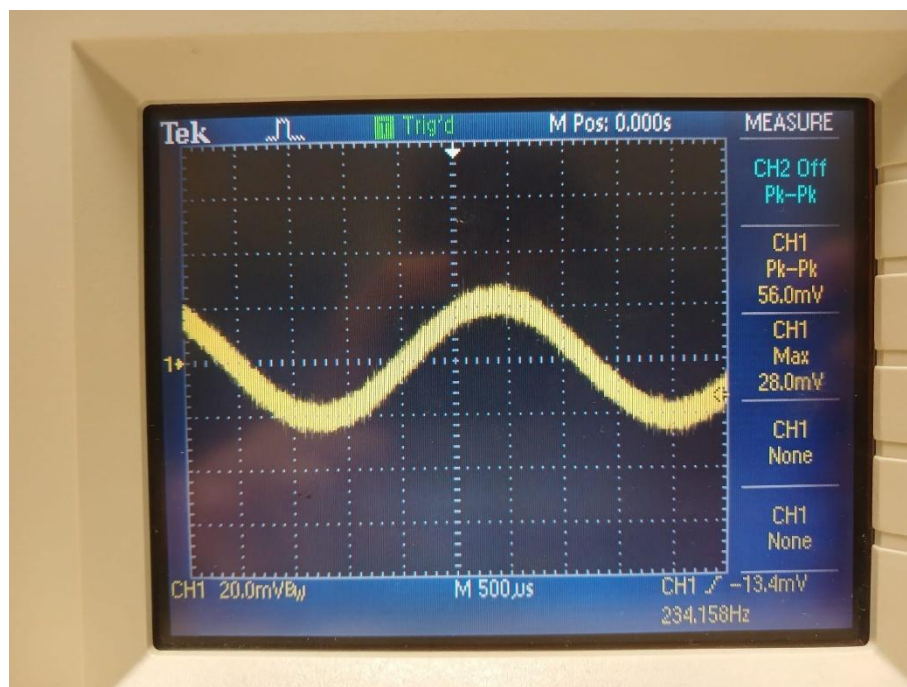


Figure 2.3: Output Voltage at $f_0/10$ with 20mV input

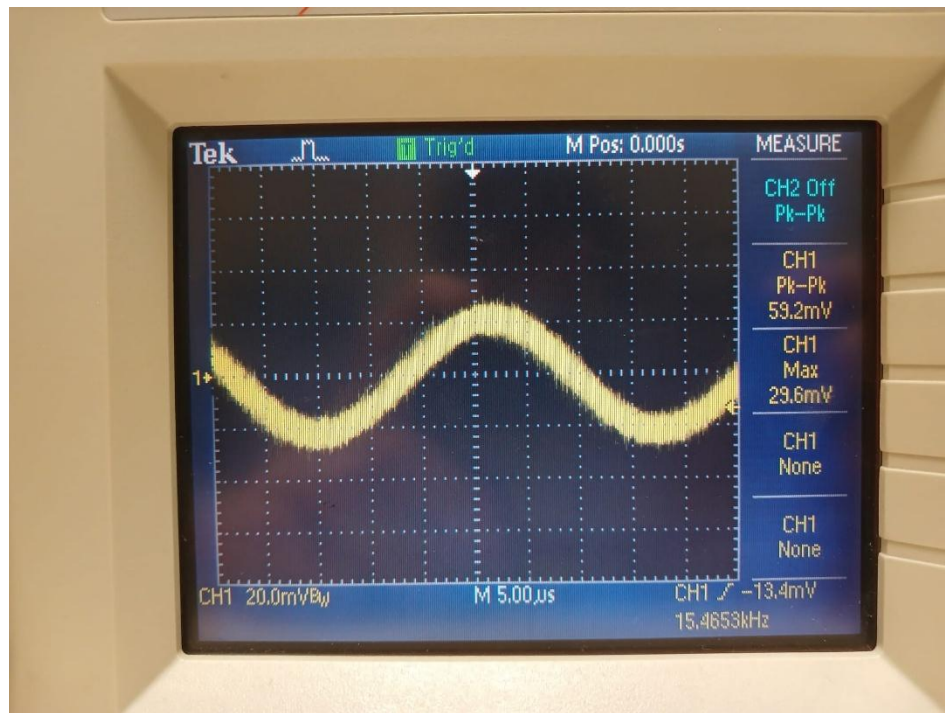


Figure 2.4: Output Voltage at $10f_0$ with 20mV input

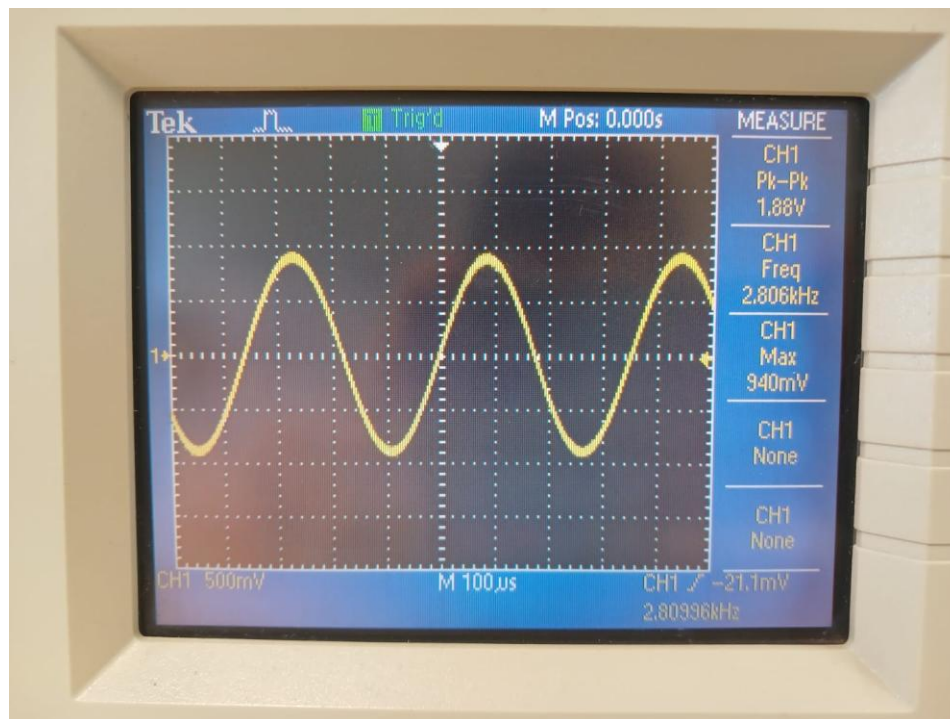


Figure 2.5: Output Voltage at corner frequency f_1 (2810 Hz) with 20mV input

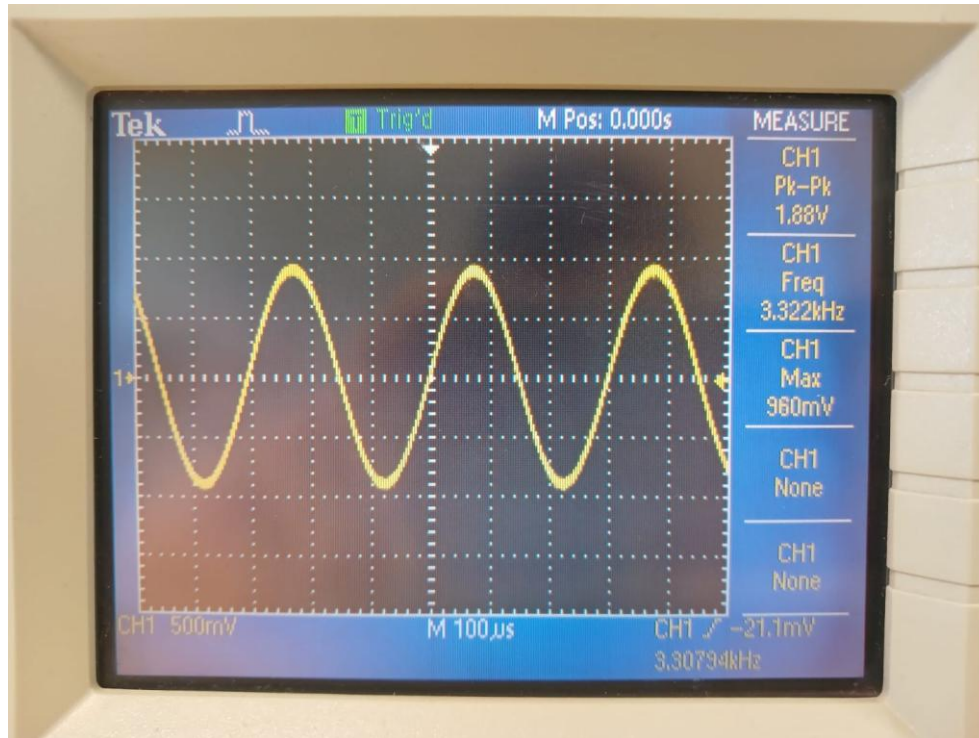


Figure 2.6: Output Voltage at corner frequency f_2 (3308 Hz) with 20mV input

f_0	Frequency (kHz)	v_{out} (V)	v_{out}/v_{in}	v_{out}/v_{in} (dB)
0.1	0.308	0.056	1.4	2.9226
0.2	0.616	0.12	3	9.5424
0.4	1.232	0.228	5.7	15.117
0.8	2.464	0.896	22.4	27.005
0.9	2.772	1.51	37.75	31.538
0.95	2.926	2.26	56.5	35.041
1	3.08	2.66	66.5	36.391
1.05	3.234	2.26	56.5	35.041
1.1	3.388	1.68	42	32.465
1.2	3.696	1.06	26.5	28.465
1.8	5.544	0.362	9.05	19.133
2.5	7.7	0.226	5.65	15.041
5	15.4	0.118	2.95	9.3964
10	30.8	0.0592	1.48	3.4052

Table 3: Measured Data for the active bandpass filter at given frequencies

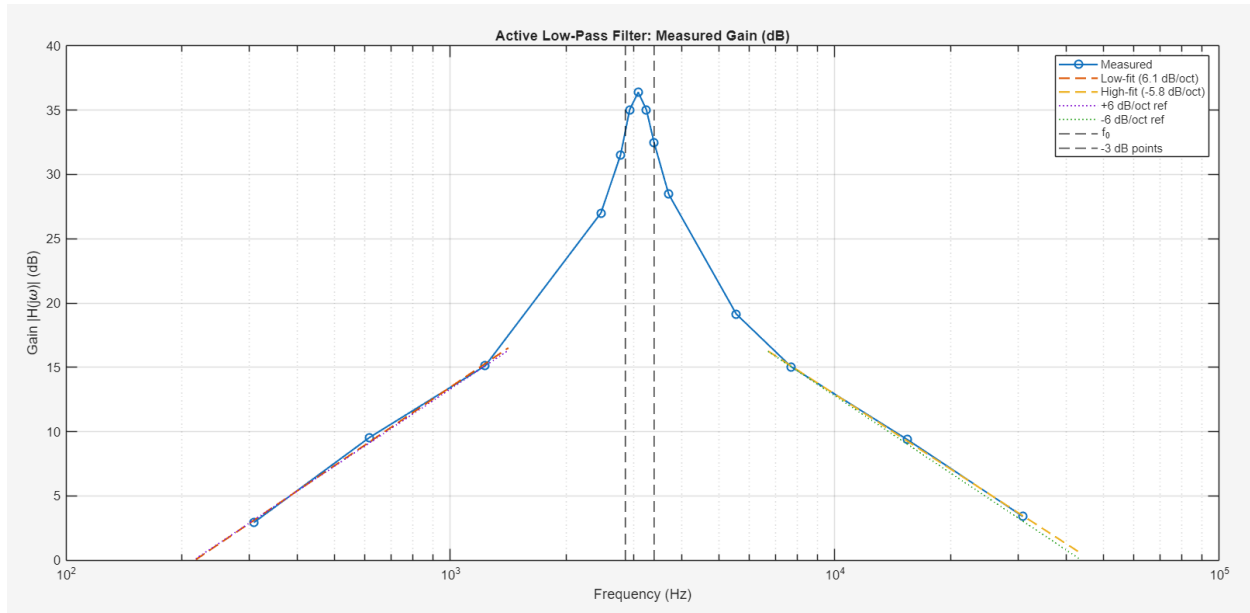


Figure 2.7: Measured gain versus frequency using the data from Table 3 (Code from Moodle)

Using these values, we were able to compare the real-life values to our simulations. First, the maximum gain at the center frequency was found to be 36.391 dB and the expected value was 36.5463 dB, yielding an error of 0.4249%. In addition to the gain, the frequencies around the center frequencies were analyzed to find the -3 dB corner frequencies of 2810 Hz and 3308 Hz, which yielded a bandwidth of 0.4980 kHz. Compared to the expected bandwidth of 0.5058 kHz, an error of 1.5663% was observed. These can be explained by the components' tolerances and the center frequency being shifted by a small amount

For the low and high frequency roll-offs, the first and third frequencies from the start and the end were used to get the slopes respectively. Using the relation of $\frac{dB_2 - dB_1}{\log_2(\frac{f_2}{f_1})}$ with the frequencies $0.1f_0$ and $0.4f_0$ for

the low frequency response was used, yielding a value of 6.0972dB/octave. When compared with the expected curve from the LTSpice plot with the value 6.2886dB/octave, an error of 3.0436%. Doing the same thing for the high frequency response with the frequencies of $2.5f_0$ and $10f_0$, a measured value of -5.8179dB/octave was observed. Comparing with the expected result of -6.2727dB/octave, an error of 7.2505% was observed. These errors could be explained by the components not being ideal and the center frequency having an offset. All errors for the active bandpass filter were put on a table for easy reading (Table 4).

	Expected	Measured	Error
Center Frequency	2.973 kHz	3.080 kHz	3.5991%
Maximum Gain	36.5463 dB	36.391 dB	0.4249%
Bandwidth	0.5058 kHz	0.4980 kHz	1.5663%
Low-Freq Response	6.2886dB/octave	6.0972dB/octave	3.0436%
High-Freq Response	-6.2727dB/octave	-5.8179dB/octave	7.2505%

Table 4: Errors for the measured vs. expected values for the active bandpass filter

Conclusion:

The purpose of this lab assignment was to design and simulate a passive third-order low-pass Butterworth filter and an active bandpass filter. First, the assigned component and design constraints were observed and related derivations were made through them. After that, the focus was shifted to the low-pass filter, as the design was made by following the aforementioned design constraints. The analytical results of the design and its response to various frequencies were plotted using MATLAB, and was simulated using LTSpice. From these plots, the cutoff frequency and the high-frequency roll-off values were measured. After the low-pass part was done, the focus was shifted to the active bandpass filter. For ease of readability, the transfer function for the given circuit was derived by Kirchhoff's Current Law for differing nodes. The schematic given in the assignment document was followed and component values were determined. These values were used with the transfer function to derive the center frequency and gain for the circuit. The gathered values were used to form the circuit in LTSpice and plot the output response for a wide range of frequencies. From the plot; the center frequency, maximum gain, the bandwidth and roll-off responses were measured. A small error for each analytical and simulative result was found. Due to the small nature of the errors, which for most cases could be explained by floating point errors, the results gathered were judged to be acceptable. From these results, we have gathered a strong basis for comparing our experimental results to the ideal cases for passive low-pass and active bandpass filters.

After the preliminary section, the derived designs were implemented in real-life using assigned components on a breadboard for the low-pass and bandpass sections. Certain measurements were carried out to compare the real results to the simulative results. A varying number of errors were observed for each value and task, whose causes were theorized under their respective calculations. Through this laboratory assignment, filter responses for passive low-pass and active-bandpass type filters were more deeply understood. Additionally, hardware implementation and circuit design skills were gained and improved.

References:

- R. E. Thomas, A. J. Rosa, and G. J. Toussaint, *The Analysis and Design of Linear Circuits*. Hoboken, NJ, USA: Wiley.
- H. Koymen and A. Atalar, *Analog Electronics*, ver. 4.2. Ankara, Turkey: Bilkent University, 2015.

Appendix:

Code for Low-pass MATLAB Plot (Figure C):

```
f0 = 629e3;
```

```
n = 3;
```

```
f = logspace(log10(6.2907e3), log10(62.907e6), 2501); % 6.2907 kHz to 62.907 MHz
```

```
H_dB = -10*log10(1 + (f./f0).^(2*n));
```

```
semilogx(f,H_dB,'LineWidth',2);  
grid on;  
xlabel('Frequency (Hz)');  
ylabel('|Vout/Vs| (dB)');  
title('3rd Order Butterworth Low-Pass Filter');  
xlim("tight")
```